

Feasibility of implementing Point-of-care testing devices for non-communicable diseases at primary healthcare level in Perú

Lack of access to laboratory tests for non-communicable diseases (NCDs) represents a major public health problem in low- and middle-income countries (LMICs). Laboratory testing is essential for the detection and management of health conditions, such as NCDs. Without these tests, the capacity to accurately monitor glucose levels, glycated hemoglobin, lipid profile and creatinine is reduced, impeding effective patient management. However, in many LMICs, patients often face challenges accessing laboratory tests at healthcare facilities. This leads to out-of-pocket expenses and, in some cases, patients skip tests, increasing the risk of treatment interruption.

This challenge is particularly evident in Peru, especially in regions such as Tumbes where diabetes prevalence (10.3%) and hypertension (26.9%) rates far exceed national averages (7.0% and 19.7% respectively). Rural and resource-limited areas face the greatest barriers to laboratory access, where infrastructure is inadequate and patients must often travel long distances for testing.

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What is a POCT device?

Point-of-care testing (POCT) devices are portable equipment that can deliver rapid and reliable results. They provide healthcare professionals with a clinically important advantage in patient management by facilitating timely and informed decisions regarding diagnoses and treatment.





Use of POCT devices for NCDs

POCT devices can provide faster, accessible and more accurate diagnoses of NCDs, particularly in resource-limited settings. They can test single parameters related to metabolic conditions (e.g., glucose or lipid profile) one at a time, or multiple hematological parameters simultaneously through multiplex cartridges, as in multiparameter POCT devices.

Implementation approach

A research project conducted in Peru determined the potential utility of a co-created strategy for implementing multiparameter POCT devices to improve care for NCDs at public primary healthcare centres in urban and rural areas. The study compared the standard of care (conventional testing) with the use of multiparameter POCT devices in the screening and monitoring of NCDs (i.e., type 2 diabetes, kidney disease and hypercholesterolemia). The intervention consisted of training for health professionals, the implementation of a POCT device in the laboratory area for a three-month period, and advertising to raise awareness of the availability of tests within the local population. The intervention also included a three-month follow-up to monitor participants.

Why is a co-created strategy?

"Incorporating the perspectives of stakeholders—including patients, communities, and health professionals—facilitates designing interventions tailored to specific health service needs, ultimately enhancing patient-level outcomes and promoting more effective healthcare delivery."

The study focused on two main outcomes: the number of diagnoses, and the number of participants who achieved their treatment goals. In addition, the study included an economic evaluation and a process evaluation to assess the cost implications, feasibility, and acceptability of implementing multiparameter POCT devices in primary healthcare centres. The main findings from these components are presented in this policy brief.

Findings

A total of 1,234 participants were recruited in eight primary healthcare facilities, including four intervention sites equipped with POCT devices and four control sites using conventional laboratory testing, in both urban and rural areas of Tumbes.

Key results

Detection of new cases of diabetes and high cholesterol increased significantly with POCT implementation compared to conventional testing, with consistent results in both urban and rural areas.

- **37.8 %** of participants received a new diagnosis of at least one NCD, such as type 2 diabetes, kidney disease, or hypercholesterolemia.
- The detection rates for diabetes and high cholesterol were higher with POCT devices, both when used alone and in combination with conventional testing.

Economic analysis

• Individuals rarely incur out-of-pocket expenses for laboratory tests, as costs are covered by the National Comprehensive Insurance scheme. Those who use intervention facilities (with multi-parameter POCT) in urban areas spent more money on lab tests than their counterparts, possibly due to increased awareness resulting from the intervention.

• Implementing multiparameter POCT devices in primary healthcare facilities is more expensive than standard care, mainly due to the costs associated with equipment procurement and staff training. Economies of scale result in lower unit costs in urban areas compared to rural areas.

Additional findings

Interviews and observations in primary healthcare facilities revealed key insights into POCT implementation and patient care delivery. Laboratory staff and healthcare workers valued the speed and comfort provided by the device but raised concerns about test accuracy due to small sample volumes. Additional challenges, including the need for training and unfamiliarity with the device were also reported. The study further examined barriers and challenges of conventional laboratory tests. Findings are summarized across four main themes:

Individuals

• Patient acceptance varied: POCT was valued for its speed and convenience. However, concerns about accuracy due to small sample volumes led some patients to prefer traditional testing, while other recognized POCT as a viable option for rapid diagnostics.

Intervention characteristics

• Capacity limitations in urban areas: Laboratory personnel valued POCT's practicality in emergencies. However, its four-sample capacity reduced efficiency during periods of high demand, increasing wait times in urban healthcare facilities.

• Higher efficiency in rural areas: In rural settings, operations were reported to run more smoothly, consistent with lower patient volumes.

Context - Inner setting

• Infrastructure deficiencies: Lack of minimum temperature control and the absence of maintenance plans for laboratory equipment were identified across facilities.

• Staff training limitations: Most laboratory personnel had only technical training suited for non-complex procedures, leading to inadequate performance of more complex analyses such as glycated hemoglobin.

• Quality control gaps: Significant deficiencies were noted in the calibration and quality control of conventional tests, worsened by frequent shortages of reagents and supplies, which hindered the establishment of effective quality systems. These issues likely contributed to discrepancies between conventional testing and POCT results.

Process

• Training builds confidence: Continuous training and support for laboratory personnel enhance their confidence in using POCT devices.

• Effective communication facilitates patient acceptance: Clear explanation of POCT's advantages by staff to patients help mitigate skepticism towards a small, rapid device that requires minimal sample volumes, especially when compared with conventional tests, which are more familiar and trusted by patients.

Lessons for POCT implementation



INFRASTRUCTURE GAPS

- POCT offered the advantage of simpler calibration and quality control.
- There was a clear disconnect and insufficient budget planning to prevent stock-outs of supplies and to ensure adquate equipment maintenance.



STAFF CAPACITY

- Laboratory personnel had limited technical training for complex tests.
- Ongoing training improved staff confidence.
- Clear communication reduced patient skepticism.
- Insufficient training to ensure proper calibration and quality of the analytical process.



DEVICE PERFORMANCE

- Speed and practicality were valued by staff.
- The four-sample capacity led to wait times in urban centers.
- Service delivery ran more smoothly in rural areas with lower patient volumes.

Challenges and limitations

- The quasi-experimental design without randomization may have introduced selection bias, especially since two intervention sites participated in the co-creation phase.
- Different conventional laboratory devices were used across control and intervention sites, which may have contributed to variation in detection rates.
- Low follow-up rates (only 20.2% of participants returned) limited the assessment of POCT's sustained impact, particularly in urban areas with high patient demand.
- The three-month intervention period may have been insufficient to evaluate long-term adherence, sustainability, and system-level changes.

For the Ministry of Health and regional health authorities:

Recommendations

1. The implementation of POCT devices for the early detection of NCDs in primary care settings should be conducted with a special focus in rural settings to close the gap of access to laboratory services.

2. Develop and implement national guidelines for POCT devices and conventional laboratories including calibration processes, and quality control, all of which are essential for ensuring accurate and precise results.

3. Establish a system to prevent laboratory supply shortages considering the number of people with chronic conditions, the periods with more demand as well as the number of tests per year that patients need.

4. Conduct regular training for laboratory personnel on good laboratory practices, use and calibration of available devices, and quality control.

"All the healthcare facilities with laboratories should have available laboratory tests to identify and control chronic diseases"

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